FORMATIVE ASSIGNMENT

Kenza Health: Early Malaria Monitoring Dashboard

Description of the concept of the Mission and solution:

Kenza Health's mission is to protect vulnerable children (0 to 5), from severe malaria in the rural southern regions of Benin, a place that serves as an incubator of *Anapheles* mosquitoes, known carriers of malaria, due to the two-Season Rain Cycle and an environment surrounded by rivers and lakes (Eg: The Mono River and Lake Ahémé) (Adigoun, n.d.; Balmith et al., 2024). Despite the fact that international organisations are providing the necessary community health workers in those regions, children are still dying from severe cases which is what Kenza Health is solving (Malaria 2024 Benin Country Profile, n.d.).

I survived a severe malaria attack in Lokossa, I experienced firsthand how the disease built up silently for days, and observed mothers and children waiting on overcrowded hospital floors because the fever was caught too late. Malaria often builds up silently, and by the time parents see severe symptoms like high fever or yellowing, it is already a medical emergency.

As prevention is one of the best tools we have against Malaria, Kenza Health uses accessible software technology (Python, SQL, a React.js Dashboard, and an SMS Gateway API) to increase reach to at-risk children before it gets too late and it relies on a hardware prototype: a wristband that detects fever and sends offline SMS alert using a MAX30205 (sensor tool specifically designed for clinical medical wearables) and a SIM800L GSM/GPRS Module (tool that connects to the standard 2G cellular networks. It can takes the data from the sensor and fires off the SMS text message to both the mother and our central software dashboard. It is cheap and tiny)

By tracking the child's body temperature at regular intervals, It detects the danger using a hard logic trigger: when a fever spike goes over 38.5 C (infection ….). When detected, it alerts the community health workers software dashboard so that they provide the necessary care as soon as possible. This software side of kenza health at the heart of this project proposal and will be fully explored in the next few pages.

However, because there is not always reliable Wi-Fi or electricity in these remote regions, the device works offline using basic GSM cellular technology to send a simple text message. The Python backend database reads this data and sends an immediate SMS alert to the mother's basic phone to take the necessary care as fast as possible. That is the hardware side that requires low level technical knowledge.

Problem statement:

Malaria remains one of the biggest challenges in the African Health sector, bearing 95% of all world cases (265 million) and 95% (579 000) of malaria deaths (*Fact Sheet about Malaria*, n.d.-a). The continent carries this burden mainly because the victims are getting the necessary care when the cases are already severe. This comes from the fact malaria progresses quickly especially in children aged from 0 to 5 as their body is not fully developed to resist the infection.

While other African countries like Nigeria account for the highest volume of cases due to the size of their population, Benin stands first as having the highest risk per person (incidence rate per 1,000 population) with 383 cases per 1,000 people (Malaria Rate by Country 2026, n.d.). Local NGOs in southern Benin like Plan International or Clinton Access Health try to protect children under 5 through community health workers. Community health workers are medical residents of those areas and provide the necessary care when needed on site. However, this does not solve the problem as the child mortality rate due to malaria stays at approximately 106 deaths per 100,000 children (in public health, this is considered as devastatingly high) due to severe cases (Benin Seeks Stronger Response to Malaria With New National Agency - Ecofin Agency, n.d.; Fact Sheet about Malaria, n.d.-b). A severe case of malaria comes from the fact that it is completely unnoticed as the severe complications only come during the first 24 to 72 hours of infection if the disease is left untreated (Trampuz et al., 2003).

Thankfully, tech solutions have been built around that same issue in eastern Africa such as NeoPanda Neoguard which is a wireless, wearable (usually on the neck or wrist on newborns) vital signs monitor that gives in real time patient monitoring and relies on bluetooth and wifi (Neopenda | Medical Technology, n.d.). While it's effective in clinical settings, this becomes unhandy in remote areas where internet data and network can be a real bottleneck.

Software development model

Kenza Health uses two distinctive ways of solving the issue of severe malaria. One through the software exclusively built for the community health workers to track the children's health in their areas and another one which directly allows the mothers to track their babies health through offline methods. As it is bi-dimensional, a rigid, linear model would be too risky. For instance, if the Python backend does not correctly format the incoming data to communicate with the React.js frontend, the dashboard would crash and children will develop the infection and die.

As it is risky, Kenza Health uses an **agile Iterative model**, where we build a whole "draft" of the prototype: (Hardware sends a basic text -> Python catches it -> Dashboard prints a raw line of text) just to prove the connection works, and then we make the dashboard look pretty.

The hypothesis

If Kenza health is deployed through a pilot program in lokossa, Benin, we expect that the number of late stage malaria deaths will reduce because the system reduces the symptom detection time from days to minutes. This is going to be evaluated through a 3 months pilot program where we are going to compare the number of late stage malaria deaths over the months. We are expecting a decrease.

To ensure this hypothesis is valid for a software deployment, it follows the SMART framework:

Specific: The pilot exclusively monitors children aged 0 to 5 in a specific rural location (Lokossa, Benin) using a defined >38.5°C fever threshold trigger.

Measurable: We will measure success by tracking the reduction in response time. We anticipate reducing the baseline detection delay from 48-72 hours down to under 2 hours, tracking how many SMS alerts result in immediate medical treatment.

Achievable: Because the hardware relies on existing, widespread 2G GSM cellular networks rather than requiring new Wi-Fi infrastructure in the villages, deploying the system is highly feasible.

Realistic: The software does not attempt to cure malaria or replace doctors; it realistically acts only as an early-warning bridge between isolated mothers and Community Health Workers.

Timely: The software dashboard prototype and the database will be built within this trimester, allowing us to focus on low level classes to build the hardware prototype.

Kenza Health

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Version 1.0

Table of content

Revision History

Name	Date	Reason For Changes	Version
Menes Adisso	May 26th 2026	First draft	1.0
-	-	-	-
-	-	-	-
-	-	-	-
-	-	-	-

1. Introduction

1.1 Purpose

The purpose of this document is to cover the software requirements of the first version (1.0) of Kenza Health. The children's health data are gathered by the hardware wristband which then saves lives through a software logic that this Software Requirements Specification (SRS) covers.

This includes the python backend logic required to receive those data from through GSM SIM 800L which then do two jobs: first, using a script, it converts the data into an SMS message, to be sent back to the mother's cellphone using the SIM 800L through a gateway. Second, it writes the data into an SQL database to render into our React,js dashboard for the Community Health Worker for them to have a clear view of the situation allowing them to take action fast.

While the physical hardware wristband generates the temperature data, this SRS focuses exclusively on those software subsystems responsible for receiving, analyzing, routing, and displaying those critical health alerts.

1.2 Document Conventions

This document follows a regular typographic convention with specification made by the African Leadership university and standard for research paper:

- Font: Times New Roman
- Font size: 11 for normal text and 14 for headings
- Line spacing: 1.5 for easier reading
- Bold wording only for highly practical terms
- APA 7th referencing
- Tables for listing technical requirements
- Diagrams for architectural components

1.3 Intended Audience and Reading Suggestions

The following serves as a technical blueprint for Kenza Health malaria monitoring system. It is mainly intended for NGOs Investors and funders who are invested into the system's main impact, cost benefit and scalability as well as Medical specialists to evaluate the effectiveness of the proactive triage approach (section 2 and 5). The primary end-users who need to understand the system's dashboard workflow, alert clarity, and the triage process (Section 3) and the technical architects (section 4).

1.4 Product Scope

The software system starts collecting the children's health data through the SIM 800L module to the python backend logic. This will allow firsthand to convert the data into an SMS message, to be sent back to the mother's cellphone allowing her to take immediate actions from exposing her child to the risk of catching a severe malaria. This is done offline, thanks to the same SIM800L module that uses a 2G GSM cellular network rather than requiring new Wi-Fi infrastructure.

Additionally, it saves the data into an SQL database to render it into an UI dashboard for the Community Health Worker for them to have a clear view of the situation allowing them to take action fast.

The software requirements and implementation can be done without high material cost as it is mainly pure coding and testing. The long-term system deployment on the other hand, requires continuous operational funding. It will rely on cloud hosting, SQL database maintenance, and recurring SMS Gateway API charges (per message sent). To maximize the accessibility of the project to vulnerable populations, Kenza Health operates strictly on a **B2B (Business-to-Business) / B2G (Business-to-Government)** procurement model. The end-users (mothers and children) will receive the hardware wristbands and SMS alerts completely for free of charge. Financial sustainability, system maintenance, and operational costs will be fully funded through pitching competitions, international health NGOs (e.g., Plan International, Clinton Health Access Initiative), and partnerships with Benin's national health agencies.

2. Overall Description

2.1 Product Perspective

Kenza Health is a new, self-contained, bi-dimensional health monitoring ecosystem designed specifically for low-resource infrastructure environments. It consists of two main interconnected components: the low cost hardware which serves as the child wristband and the cloudware which contains the software system. This SRS specifically defines the functional and non-functional requirements of the software subsystem.

It connects directly with the hardware device that the child wears on their wrist. When the MAX30205 sensor on the wristband detects a high temperature, the SIM800L module uses the local 2G network to send a raw SMS text payload. Our Python backend script then catches this message, reads the data, and does two jobs at the same time: it fires an automated warning text to the mother's basic phone through an SMS Gateway API, and it saves the records in an SQL database so the Community Health Workers can see the emergency live on their React.js dashboard.

2.2 Product Functions

- Reading the wristband data (continuously listens for incoming text)
- Fever checks and alert (>38.5C emergency triggered)
- Data persistence logging onto SQL
- Real time dashboard

2.3 User Classes and Characteristics

User groups	Characteristics	Technical needs
Children under 5	Those are the **primary user** target, children aged from 0 to 5, as their body is not fully developed, they are the primary victims of Malaria	The wristband with the sensors that monitors their temperature at night
Parents/Guardians/Mother	They have a low technical literacy and are in charge of taking care of the household and the children meaning they do not have time to navigate complex applications.	In case the mother has a cellphone, they must not need a Wifi device to save their children, the safety of their children relies on the SMS notification. In case they don't, the community health worker takes it.
Community health workers	Those are medical residents that live directly in the region. They provide the necessary care when needed.	Need an intuitive, and smooth dashboard to be able to easily track the children in those villages.

2.4 Operating Environment

Kenza Health will depend on 3 interconnected and operational environment:

- Server side environment (Backend and database): Will be hosted on a lightweight cloud server AWS EC2 micro running **Ubuntu Linux 22.04 LTS**. For parsing the script and converting data into an SMS message, **Python 3.10+** will be used, and an **SQL Database** (MySQL or PostgreSQL) to store the data.
- Network: The system relies entirely on local 2G GSM infrastructure provided by **MTN Benin** and **Moov Africa** to ingest inbound hardware data strings via SMS. it will connect seamlessly with a third-party cloud SMS Gateway API via secure HTTPS requests to route automated emergency texts back to the mothers' basic phones.
- Community Health workers dashboards: They will basically need to run their device onto an up to date browser (Chrome, Edge, Safari etc..).

2.5 Design and Implementation Constraints

The entire software system relies on the wristband getting first the data, which is a constraint. And designing the wristband so that it does not harm the children's health (Eg: warming excessively), lightweight despite all the components needed inside and resilient because we are dealing with youth.

Also there is the fact that the wristband will rely on its battery. Which type is the most durable, chargeable or replaceable one?

The SMS language, adapting it to the different languages spoken in the area if not French, such as (Fongbè, Minan etc..). And also tackling the cost of sending an sms through a network.

The dashboard should be able to tell the Community Health Workers which household to check out when noticing a fever.

2.6 User Documentation

To ensure a smooth and seamless adoption to Kenza Health ecosystem, community health workers will be provided a manual that explains how the dashboard works and how to navigate it.

Furthermore, a single-page, physical card given to mothers when they receive the wristband. It illustrates how to set it up and what the automated Python text messages mean and what immediate steps to take when a warning arrives.

2.7 Assumptions and Dependencies

Assumptions:

We assume that mothers will keep the wristband securely on the child (aged 0–5) continuously, and that the children will not easily damage or remove the hardware.

Basic Literacy & Tech Comfort: We assume that mothers possess at least a basic mobile phone ("dumbphone"), know how to operate a SIM card, and can read/understand simple text message alerts.

We assume that once a mother receives an emergency SMS alert, she will immediately take home-care actions and prepare to receive the Community Health Worker.

Dependencies:

2G Network Stability: The software's ability to receive telemetry completely depends on the local network uptime and 2G coverage provided by MTN Benin and Moov Africa in the Lokossa region.

Access to Power (Electricity): The entire system depends on regular access to electricity in the region so that mothers can keep their basic phones charged and the wristband batteries can be regularly replenished.

CHW Connectivity: While the mothers operate offline, the React.js dashboard depends on the Community Health Workers having at least a basic 3G/4G cellular data connection to load and update patient records.

3. External Interface Requirements

3.1 User Interfaces

The Kenza Health system only requires one GUI (Graphical User Interface) which will display real Web Dashboard busing using [React.js] essentially for the Community Health Workers (CHWs). Neither the mothers nor the children should interact with the software interface.

It will be displayed as a responsive, two column layout. The left side should show the navigation options which are: Home (Welcoming page + overview dashboard of dataset), Active patients (Where the CHW can actually see when a child needs attention) and then the settings.
The UI must automatically adjust its grid size to display properly on low-cost Android tablets, smartphones, and laptops used in the area.

To ensure immediate triage, the patient list is automatically sorted by medical emergency using high color contrast standards:
Critical Status (Red): Used for children with a fever reading ***>38.5C***. These cards automatically indicate signals.
Normal Status (Green): Used for children whose last recorded temperature is within the safe range of 37.5.
Offline Status (Gray): Used if a device has failed to ping the Python backend for more than 6 hours, indicating a potential network issue or battery drop.

3.2 Hardware Interfaces

Our system wirelessly communicates with the Kenza wristband (MAX30205 sensor + SIM800L module).
The data flows from one point to another via 2G GSM SMS messages over local cellular networks.

3.3 Software Interfaces

Software interfaces	Description
SQL Database (PostgreSQL/MySQL)	It acts as our storage of data. The python script incoming data (Device_ID, Temp, Timestamp) into the tables that React.js will render into the dashboard.
SMS Gateway API	A third party cloud service. The python backends data to this API to trigger the emergency text.
OS and Python libraries	Running on Ubuntu Linux Python 3.10+, relying on standard libraries (like requests for the API and psycopg2/SQLAlchemy for the database)

3.4 Communications Interfaces

Communication s interfaces	Description
Cellular SMS (inbound)	The wristband sends data over the 2G network (e.g., ID:102, T:39.1)
HTTPS REST API's	The Python backend talks to the SMS Gateway API over the internet using secure HTTPS. The data messages sent there are formatted in JSON.
Security	Uses standard **TLS/SSL** protocols to protect the children's medical data (For all internet connections)
Web dashboards	The front end updates from the server using standard **HTTPS** via the health worker's web browser.

4. Requirement Specification

4.1 Stakeholder Requirements Specificµation

Functional Requirements

Req ID	Requirements	Description
FR 1	**Data ingestion**	
FR 1.1	Receive SMS	Allow the backend to receive the raw text transmissions from the SIM800L.
FR 1.2	Parse Telemetry Data	Extracting the data and separating it into clear variables.
FR 2	**Fever analysis**	
FR 2.1	Evaluating the temperature	Detecting when the body temperature is over 38.5C
FR 3	**Alert system**	
FR 3.1	Dispatch the alert to the mother	Sending automatically an urgent care message to the mother's mobile phone through the SMS Gateway API when a fever is detected
FR 4	**Data Logging**	
FR 4.1	Storing all health records	Allow the system to save permanently into our SQL database
FR 5	**CHW dashboards**	
FR 5.1	Displaying the patient grid	A real time react.js dashboard that shows all children that are monitored from the CHW
FR 5.2	Prioritizing the critical cases	Highlighting the ones that have fever in red
FR 5.3	Update the status	Allowing the CHW to know when an emergency has already been treated.

5. Other Nonfunctional Requirements

Requirement Type	Req ID	Description
Security (User Access)	NF 1	An authentication system (SHA-256) that authenticates the CHW securely before granting access to the patient dashboard transmitting through AES-128 encryption.
Safety	NF2	Rubber materials for the wristband as well as casing materials, loop to log data locally if network connection drops.
Performance	NF 3	The Python backend must be able to process inbound SMS telemetry from up to 500 active wristbands without delaying alerts at the same time
Usability	NF 4	The language has to be basically French offering the possibility to translate in Fongbe or Minan audio translation.
Availability	NF 5	The system needs to be available 24/7 to ensure that the midnight spikes are never missed.
Cross Browser Support	NF 6	The React.js dashboard must render correctly across Google Chrome, Mozilla Firefox, and Safari mobile browsers.
Technology	NF 7	The system dashboard must be fully accessible on most operating systems, on any mobile device and standard PCs operating on basic 3G/4G networks.

Business rules

The platform will only classify a child's status as red (critical) and prioritize the case on the dashboard only if MAX30205 sensor data indicates a body temperature superior to 38.5C. Once this threshold is reached, the python backend automatically sends an SMS notification to the mother and alerts the CHWs. After human review, the CHW explicitly interacts with the "mark as treated" button on the dashboard.

6. Appendix

Appendix A: Glossary

AES-128: It's an advanced Encryption Standard (128-bit); an algorithm used to secure the transmission of telemetry data over a network

API (Application Programming Interface): A set of communication rules that allows the Python backend to talk to the third-party SMS Gateway

B2B / B2G: Business to Business and Business to Government

CHW Community Health Workers

Fongbè / Minan: Local national languages spoken in the southern regions of Benin (such as Lokossa)

GSM (Global System for Mobile Communications) it's the standard digital cellular network used by mobile devices, enabling the wristband to send texts without Wi-Fi

GUI Graphical User Interface

HTTPS: Hypertext Transfer Protocol Secure; the encrypted internet protocol used for the backend to communicate with web services securely

JSON (JavaScript Object Notation): A lightweight data-formatting standard used to structure the text messages sent to the API.

MAX30205: A clinical-grade digital hardware temperature sensor

React.js: A popular software library used to build fullstack apps

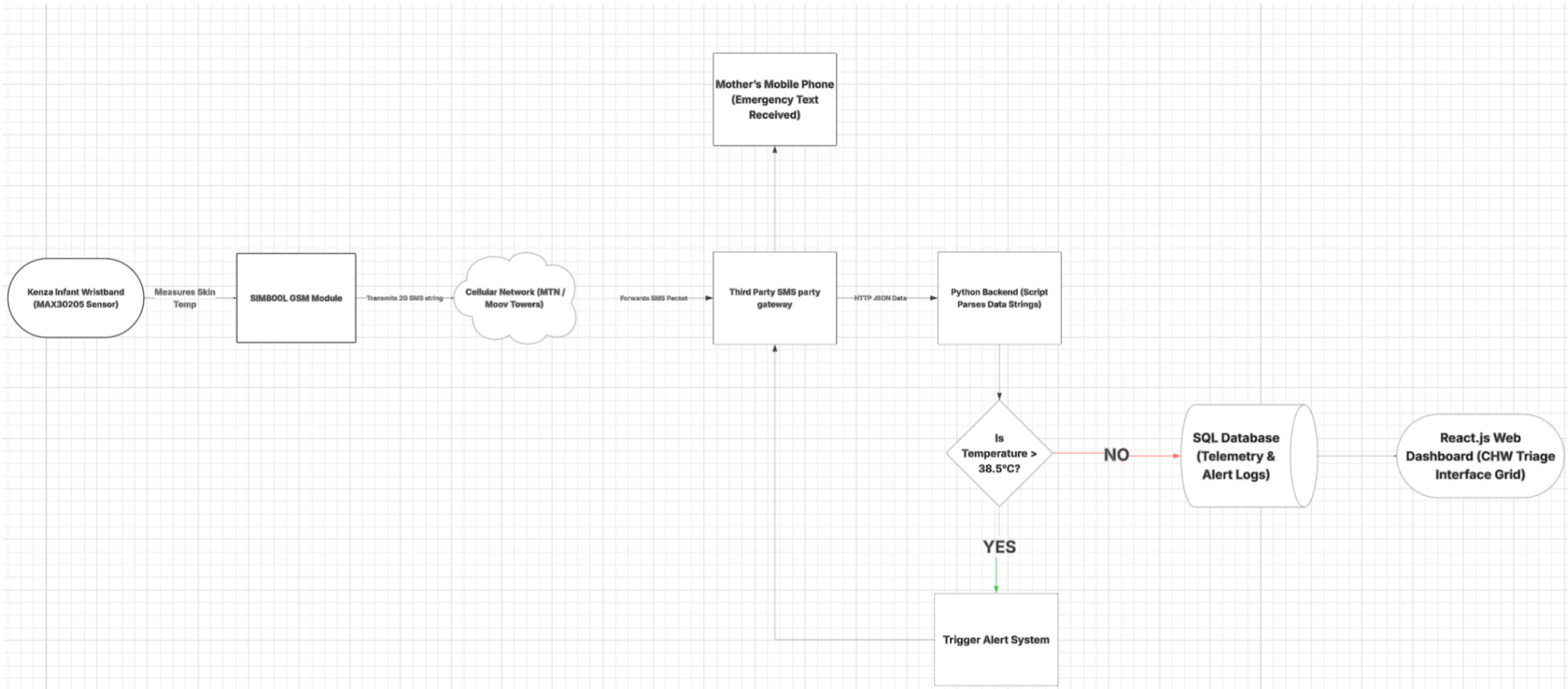
SIM800L: A very small cellular hardware module that reads the temperature sensor data and transmits it as an SMS text message.

SQL (Structured Query Language): A standard programming language used by the backend database to permanently store data

TLS / SSL: standard security protocols used to create encrypted links between the web server and the browser

Appendix B: Analysis Models

Kenza Health Flowchart



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